

tmux documentation

Changelog — vasic-digital/tmux

All releases use [Semantic Versioning](#). Every release carries a positive-runtime-evidence verification record per the project's anti-bluff covenant (Constitution §101 / universal §11.4).

[v1.0.6] — 2026-05-21

****Workable-items closure cycle:** §11.4.80 cadence wired (launchd + systemd user timer + git pre-push hook), Containers/QWEN.md covenant gap closed

- pushed to Containers remotes (fbef9d6). Full rebuild + install + quintuple verification GREEN. Release tag per §11.4.40.**

Added (project)

- `scripts/codegraph_cadence_check.sh` — §11.4.80 cadence-floor enforcement (default 7-day floor; configurable via `CODEGRAPH_CADENCE_DAYS`). Parses `.gitignore-meta/.regenerated/ codegraph-db.ok` stamp, extracts `regenerated_at + node_count`, reports GREEN / STALE / ENV with §11.4.6 honest reasons. Anti-bluff: reads stamp CONTENT (not just existence); stamp with `node_count=0` is STALE even if recent.
- `scripts/codegraph_install_cadence.sh` — §11.4.81 cross-platform cadence-trigger installer. Darwin path: writes a launchd plist at `~/Library/LaunchAgents/digital.vasic.tmux.codegraph-cadence.plist` with `StartInterval=604800` (7 days). Linux path: writes a systemd user timer + service unit at `~/.config/systemd/user/`. Cross-platform: writes git pre-push hook at `.git/hooks/pre-push` calling `codegraph_cadence_check.sh`; `CADENCE_MODE=warn` (default) prints warning + allows push; `CADENCE_MODE=block` refuses push when STALE. Idempotent install + clean --uninstall path.

Fixed (project + Containers submodule)

- **Containers/QWEN.md covenant gap** (fbef9d6 pushed to Containers github + gitlab). Inserted the verbatim 2026-04-28 user-mandate quote + §11.4.81 cross-platform-parity reference into the consumer-layer QWEN.md (the other Containers governance files — CLAUDE.md, AGENTS.md, CONSTITUTION.md, Constitution.md — already carried the covenant; QWEN.md was the only gap). Containers submodule pointer bumped 4ca5491 → fbef9d6 in this project commit.

Verification (this cycle, captured 2026-05-21 on Darwin arm64)

- `bash scripts/setup.sh` (full rebuild + install) → GREEN. `~/.tmux.conf` installed; snippet appended to `~/.bashrc` + `~/.zshrc`; tmx wrapper installed; PATH propagation verified.

- `bash scripts/setup.sh --verify-only` → SUMMARY PASS=22 FAIL=0 SKIP=2 GREEN.
- `bash scripts/tests/meta_test_false_positive_proof.sh` → 32 caught / 0 escaped / 6 skipped GREEN.
- `bash scripts/test_e2e.sh` → PASS=9 FAIL=0 SKIP=0 GREEN.
- `bash scripts/codegraph_validate.sh` → PASS=4 FAIL=0 SKIP=1 (V4 honest gap re submodule traversal — same as v1.0.5).
- `bash scripts/codegraph_cadence_check.sh` → GREEN (0.2d ago, 6 nodes, within 7d cadence floor).
- `zsh -ic 'which tmx; tmx -V'` → /Users/milosvasic/Projects/tmux/ scripts/tmx + tmux 3.6a (operator-path live confirmation).
- `launchd job loaded: launchctl list | grep codegraph-cadence` shows `digital.vasic.tmux.codegraph-cadence`.
- git pre-push hook installed + executable: `.git/hooks/pre-push`.

Hardened (4-layer regression protection per §103)

- **Layer 1 static:** cadence check parses real stamp content (not just file existence); reads codegraph CLI version + node count.
- **Layer 2 runtime:** `codegraph_cadence_check.sh` runs against the freshly-regenerated stamp every cycle (reported GREEN in this release's verification batch).
- **Layer 3 challenge:** existing TMUX-CH-20/21/22 cover the CodeGraph install + index + MCP wiring; the cadence layer is a refinement of the CH-20 install path.
- **Layer 4 mutations:** (deferred — cadence-script paired mutation is a §11.4.6 honest-tracked TODO for v1.0.7; the cadence script itself uses captured-evidence reads per §11.4.5).

§11.4.40 release-tag discipline

This release is created AFTER the complete fresh retest triple this session (verify + meta + e2e + `codegraph_validate` + cadence + tmx operator-path) — not a spot-check. All five GREEN. Tag `v1.0.6` created post-commit, pushed to github + gitlab.

§11.4.71 pre-push integrity

- Parent: at upstream tip; no divergent commits.
- `constitution/` submodule: at upstream tip (6e164f3); pulled in v1.0.5; no new commits this cycle.
- `Containers/` submodule: bumped 4ca5491 → fbe9d6 in THIS project commit; the fbe9d6 Containers commit was pushed to github + gitlab in this same session before the pointer bump.

Out-of-scope (honest tracking per §11.4.6)

- Cadence-script paired §1.1 mutation — deferred to v1.0.7. The cadence script's correctness is currently verified by its own output reading captured stamp content; a mutation that backdates the stamp and asserts the cadence-check FAILs would tighten the loop but isn't yet wired.
 - CodeGraph upstream `--include-submodules` — out of scope per §11.4.74.
 - Linux-host CI runner — would exercise the Linux branches of tests 09/13/14 + the systemd user timer install path.
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[v1.0.5] — 2026-05-21

§11.4.81 cross-platform-parity discipline landed universally + project test 09/13/14 gain Darwin branches + NEW test 24 (Darwin CPU-cap via RLIMIT_CPU+SIGXCPU) +

§11.4.79 own-org submodule inclusion fix + constitution submodule bumped

(19ce1b1→6e164f3) with the new §11.4.81 anchor universal across every consuming project.

Added (constitution submodule, pushed 6e164f3)

- **§11.4.81 — Cross-platform-parity mandate.** Universal anchor + mirror blocks in `Constitution.md`, `CLAUDE.md`, `AGENTS.md`, `QWEN.md`. Three sub-mandates: (A) per-OS implementation REQUIRED via runtime `uname -s` dispatch, (B) per-OS tests REQUIRED with positive captured evidence per branch, (C) honest kernel-gap citation + adjacent equivalent test REQUIRED where no equivalent exists (canonical: XNU `RLIMIT_AS` unprivileged → use `RLIMIT_CPU+SIGXCPU` adjacent). Per-OS equivalence catalogue listed. Composes with §11.4.1/2/3/4/5/6/20/27/ 69/70 + §107. Pre-build gate `CM-CROSS-PLATFORM-PARITY` planned; paired §1.1 mutation: strip a Darwin branch → gate FAILs.
- **Constitution submodule QWEN.md** gained the verbatim 2026-04-28 anti-bluff user-mandate quote (audit identified it was missing there — every consumer file at every layer now carries the literal).

Added (project)

- **NEW** `scripts/tests/24_cpu_cap_enforcement.sh` — Darwin `RLIMIT_CPU` + `SIGXCPU` enforcement test. Per §11.4.81 (C) the adjacent test for what Linux tests via `cgroup MemoryMax` (test 12). Captures: process killed by signal 24 (`SIGXCPU`) after ~3s wall given `ulimit -t 2`; `TMX_CPU_HARD_SEC=7200` propagates to `RLIMIT_CPU=7200` inside the session.
- **NEW** `scripts/codegraph_validate.sh` — §11.4.78 step 4 + §11.4.79 validate-probe + §11.4.80 sync-script callee. 5 probes: V1 CLI version, V2 node count > 0, V3 §11.4.79 own-org/third-party split, V4 honest-gap re submodule traversal (SKIP not FAIL), V5 MCP server spawn smoke.

Fixed (project — §11.4.79 + cross-platform parity)

- **A26 — §11.4.79 compliance: own-org submodules removed from CodeGraph exclude.** v1.0.4's `.codegraph/config.json` was a §11.4.79 violation: `constitution/**` + `Containers/**` were excluded. v1.0.5 removes them from `exclude` (own-org MUST be INCLUDED); keeps `tmux/**` excluded (third-party). Honest gap: CodeGraph 0.6.8 does not traverse git submodules from the parent index; config compliance is met; actual cross-submodule indexing waits for upstream CodeGraph - `- include-submodules` (out of scope per §11.4.74).
- **A25 — Darwin branches for Linux-only tests (§11.4.81 fix).** Tests 09/13/14 dispatch on `uname -s`:
 - **09 D-***: spawn 2 operator-path sessions; `rlimit` wrapper invoked; read `ulimit -t/-u` inside each pane; `SIGKILL` session A's server; verify B survives with ORIGINAL PID. PASS=6/0/0.

- **13 D-***: `RLIMIT_NPROC` fork-bomb probe — child `bash` lowers `ulimit -u 64`, fork-bombs; captures `EAGAIN` occurrences from `stderr` (`bash: fork: Resource temporarily unavailable = XNU kernel-enforced`). `PASS=2/0/0`.
- **14 D-***: 3 sessions A/B/C; `SIGKILL` A's server (macOS adjacent test for OOM-independence per §11.4.81 (C)); verify B+C survive with `ORIGINAL` PIDs + `tmx ls` still lists them. `PASS=5/0/0`.
- **A24 — Constitution submodule pointer bumped** (19ce1b1→6e164f3) in same commit as cascade work per §11.4.26 step 7.

Hardened (4-layer regression protection per §103)

- **Layer 2:** tests 09 D-, 13 D-, 14 D-, 24 D- — all `PASS` this cycle with positive captured runtime evidence per platform branch. Suite total this cycle: `PASS=22 SKIP=2` (was 18 `PASS` / 5 `SKIP` in v1.0.4). The improvement: tests 09/13/14 dispatched to Darwin branches (was `SKIP`), test 24 `NEW`.
- **Layer 3:** `TMUX-CH-24` added; `CH-09/13/14` challenges already cover the (now multi-branch) invariants.
- **Layer 4 (paired mutations):**
 - **M7-M10 RETIRED** — targeted dead `scripts/tmx-vm` (legacy VM wrapper, replaced by native dual-OS per `Fixed.md` A4-A8). Dead-code mutations were inflating `SKIP` count without coverage signal.
 - **M20 (NEW)** — strip `ulimit -t` from Darwin `rlimit` wrapper → test 15 T5 `FAILs`. Topology-guarded: Linux uses `cgroup` (covered by M5).
 - **M21 (NEW)** — clobber `ulimit -u` to 1 in Darwin `rlimit` wrapper → session lifecycle breaks (`NPROC=1` cannot fork helpers). Honest §11.4.6 note: stripping the line wouldn't change `readback` because macOS host default `ulimit -u` happens to match wrapper's 2666; clobber-to-1 forces an observably wrong value.
 - **M22 (NEW)** — re-exclude `Containers/**` from `.codegraph/config.json` → `codegraph_validate.sh` V3 `FAILs` (§11.4.79 violation detected). Caught + restored both directions.

Meta-test summary: **32 caught / 0 escaped / 6 skipped** (M4/M5 Linux-only-topology + M7-M10 retired with explicit `SKIP-with-` rationale).

Verification (this cycle, captured 2026-05-21 on Darwin arm64)

- `bash scripts/setup.sh --verify-only` → `GREEN`; suite `PASS=19 FAIL=0 SKIP=4` (the 4 `SKIPs` are the 3 destructive-Linux tests + the `oom_score_adj` Linux-only test). Test 24 `NEW`; tests 09/13/14 now contribute Darwin `PASS` counts where they previously contributed Linux-only `SKIP`.
- `bash scripts/tests/meta_test_false_positive_proof.sh` → **32 caught / 0 escaped / 6 skipped GREEN**.
- `bash scripts/test_e2e.sh` → `GREEN`.
- `bash scripts/codegraph_validate.sh` → 4 `PASS` / 0 `FAIL` / 1 `SKIP` (the `SKIP` is V4 honest-gap on submodule traversal).
- `bash scripts/codegraph_reindex.sh` → 6 nodes, stamp written.
- §11.4.71 pre-push: parent + `constitution/` (6e164f3) + `Containers/` (4ca5491) all at upstream tip, no divergent commits.

Out-of-scope this cycle (honest tracking per §11.4.6)

- §11.4.80 automatic-trigger wiring (cron / git hook for the constitution- provided `codegraph_update.sh + codegraph_sync.sh`) — deferred to next cycle. Manual invocation works today.
 - CodeGraph upstream support for `--include-submodules` — out of scope per §11.4.74.
 - Containers/QWEN.md create — separate PR to vasic-digital/Containers per §11.4.28 owned-submodule equal-codebase mandate.
 - Linux-host CI runner — would let us exercise the Linux branches of tests 09/13/14 in addition to the Darwin branches running today.
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[v1.0.4] — 2026-05-21

CodeGraph code-intelligence integration (§11.4.78), anti-bluff covenant propagated verbatim to every consumer governance file, audit follow-up fixes (M4/M5 portability + `tmx kill shorthand`), docs reorganised under context subdirectories per the constitution rule.

Added

- **A18 — CodeGraph integration (§11.4.78).** Installed `@colbymchenry/codegraph v0.6.8` globally (npm prefix user-writable; no sudo per §11.4.78). `codegraph init + indexed`; config tracked (`.codegraph/config.json`), DB gitignored (`.codegraph/codegraph.db`). §11.4.10 secret-exclusion patterns + §11.4.28 owned-submodule paths (`constitution/**`, `Containers/**`, `tmux/**`) added to `exclude`. §11.4.77 regeneration mechanism at `.gitignore-meta/codegraph-db.yaml`
 - executable `scripts/codegraph_reindex.sh`. MCP wired for 5 CLI agents:
 - Claude Code (project-scoped `.mcp.json`, NEW)
 - OpenCode (`~/ .config/opencode/opencode.json`, pre-existing — audited)
 - Kimi CLI (`~/ .kimi/mcp.json`, pre-existing — audited)
 - Crush (`.crush.json`, NEW)
 - Qwen Code (`.qwen/settings.json`, NEW) All configs reference the bare `codegraph` command on `PATH` (no hardcoded host paths) — portable across machines.
- **A19 — Verbatim anti-bluff covenant in every consumer governance file (user mandate, 2026-05-21).** Inserted the literal 2026-04-28 user-mandate quote into project `CLAUDE.md`, `AGENTS.md`, and `QWEN.md` as a `## MANDATORY ANTI-BLUFF END-USER-QUALITY COVENANT` block directly after the inheritance pointer. Project `Constitution.md` already had it. Tools that don't expand `@imports` now still read the covenant.
- `docs/codegraph/README.md` — comprehensive CodeGraph documentation (§1-§11): install, prereqs, repo-tracked artefacts, secret-exclusion contract, per-agent MCP wiring table, anti-bluff verification, unforgeable-challenge note, operator-path examples, honest gaps, troubleshooting.
- `docs/plans/v1.0.4.md` — full working plan written at session start, then executed end-to-end this cycle.

- `scripts/export_docs.sh` — idempotent §11.4.65 universal-Markdown export wrapper (pandoc HTML + weasyprint PDF, per-file timeout 60s, ≤500 candidates).

Fixed

- **A20 — M4/M5 paired mutations honest topology dispatch (AUDIT-1).** Pre-fix: raw GNU `sed -i` silently SKIPPed on Darwin BSD `sed` with the wrong reason ("mutation command failed to apply"). Root cause: not just `sed` portability — the mutations target the Linux `cgroup/ systemd-run` code path of the wrapper, which Darwin doesn't reach (native dual-OS uses POSIX `rlimit` instead). Fix: explicit `uname -s` topology guard around M4/M5 — SKIP-with-reason on non-Linux per §11.4.3; portable `inplace_sed` on Linux.
- **A21 — `tmx kill` shorthand resolves to `kill-session` (AUDIT-2).** README/AGENTS commands table lists `tmx {new|attach|ls|kill}` as friendly verbs. Pre-fix: bare `tmx kill -t NAME` was passed through to `tmux` which rejected it as ambiguous (could be `kill-pane` / `-server` / `-session` / `-window`). Fix: SUBCMD-translation hook in `scripts/tmx.template` detects bare `kill` and rewrites "\$@" to use `kill-session`.

Hardened (4-layer regression protection per §103)

- **Layer 1 (static gate):** `scripts/verify.sh` gained a second Layer-1 block greppin g each of the 4 consumer governance files for the literal verbatim-covenant anchor; pre-suite refusal if any is missing.
- **Layer 2 (runtime, operator-path per §102):** 5 new tests, all PASS:
 - `19_covenant_propagation.sh` (PASS=7/0/0)
 - `20_codegraph_installed.sh` (PASS=5/0/0)
 - `21_codegraph_index_present.sh` (PASS=4/0/0)
 - `22_codegraph_mcp_wired.sh` (PASS=7/0/0)
 - `23_tmx_kill_shorthand.sh` (PASS=5/0/0)
- **Layer 3 (Challenges):** TMUX-CH-19 through TMUX-CH-23 in `scripts/challenges/tmux.yaml`.
- **Layer 4 (paired mutations):** 5 new in the meta-test:
 - M15 strip covenant (TEMP COPY of CLAUDE.md — real file untouched)
 - M16 strip `**/*.pem` from `.codegraph/config.json`
 - M17 strip codegraph from `.mcp.json`
 - M19 strip AUDIT-2 block from `scripts/tmx`
 - M4/M5 topology guard (the meta-test itself is layer 4 — pattern closes the long-standing latent BSD-sed bluff)

Reorganised

Docs moved under context-named subdirectories per the constitution rule (the user mandate 2026-05-21 explicitly invoked this):

- `docs/GUIDE.md` → `docs/guide/README.md`
- `docs/SCROLLING.md` → `docs/scrolling/README.md`
- `docs/CODEGRAPH.md` → `docs/codegraph/README.md`
- `docs/CONTAINERIZATION_PLAN.md` → `docs/plans/containerization.md`
- `docs/NATIVE_DUAL_OS_PLAN.md` → `docs/plans/native-dual-os.md`
- `docs/PER_SESSION_ISOLATION_PLAN.md` → `docs/plans/per-session-isolation.md`

- docs/PLAN_v1.0.4.md → docs/plans/v1.0.4.md

Every reference across the codebase + governance files updated atomically.

Verification (this cycle, captured 2026-05-21 on Darwin arm64)

- bash scripts/setup.sh --verify-only → GREEN; suite PASS=18 FAIL=0 SKIP=5 (SKIPs all Linux-only/destructive).
- bash scripts/tests/meta_test_false_positive_proof.sh → 26 caught / 0 escaped / 6 skipped GREEN. The 6 SKIPs are §11.4.3 topology-correct (M4/M5/M7/M8/M9/M10 — Linux-only mutations).
- bash scripts/test_e2e.sh → GREEN.
- bash scripts/codegraph_reindex.sh → 6 nodes, stamp written.
- §11.4.65 export-sync: every consumer Markdown has fresh HTML + PDF siblings via scripts/export_docs.sh.
- §11.4.71 pre-push: parent + constitution/ + Containers/ all at upstream tip; no divergent commits.

Out-of-scope this cycle (honest tracking per §11.4.6)

- Upstream constitution/QWEN.md covenant insert — needs separate PR to HelixDevelopment/HelixConstitution. The operator directive 2026-05-21 explicitly forbids modifying constitution from inside this project.
- Containers/QWEN.md create — needs separate PR to vasic-digital/Containers (its own §11.4.28 owned-submodule cycle).
- Shell parser for CodeGraph — upstream contribution to add tree-sitter shell would lift the node count from 6 to ~3000. Out of scope per §11.4.74; tracked at the upstream project.
- Agent-driven unforgeable-challenge end-to-end test — classified AUTONOMOUS_DESIGNED per §11.4.52 carve-out (mechanical seam exists via test 22 T7; agent-driven layer lands when a headless agent harness is wired).

[v1.0.3] — 2026-05-21

tmux scrolling fixed for the Claude Code TUI and mobile (Termux); governance refactored to inherit from the HelixConstitution submodule.

Fixed

- **A16 — Scrolling terminal output up/down did not work, especially in the Claude Code TUI.** Two root causes: (1) the history-limit default (2000) was too small, and (2) tmux's default WheelUpPane binding forwards the wheel to applications that request mouse reporting (Claude Code, vim, less) — so the wheel never reached tmux's own scrollbar buffer. scripts/tmux.conf.template now:
 - bumps history-limit to **50000**;
 - sets mode-keys vi for vi-style copy-mode navigation;
 - overrides WheelUpPane / WheelDownPane so the wheel and touch-scroll **always** drive tmux copy-mode scrollbar — working identically on a desktop mouse, a trackpad, and a phone (Termux/Android touch-scroll → wheel events);
 - adds the official Claude Code passthrough settings (allow-passthrough on, extended-keys on, terminal-features 'xterm*:extkeys') so Shift+Enter and escape sequences reach the application;

- adds OS-adaptive clipboard routing (pbcopy / wl-copy / xclip / termux-clipboard-set, detected at copy time) plus OSC-52.

Added

- **A17 — HelixConstitution governance submodule + verified inheritance.** The universal engineering rules (anti-bluff covenant, data safety, memory budget, continuation invariant) now live in the HelixDevelopment/HelixConstitution submodule at constitution/ (pinned 7f738df). The project's Constitution.md was refactored to the extends-template form (Project Articles §101–§109); CLAUDE.md / AGENTS.md gained INHERITED-FROM pointer blocks; a new QWEN.md was added for the Qwen Code CLI agent. The Containers submodule is HelixConstitution-wired too (recursive inheritance via find_constitution.sh).

Hardened (4-layer regression protection per Constitution §103)

- **Layer 1 (static gate):** scripts/verify.sh gained a pre-suite static gate that greps tmux.conf.template for every scroll setting; RED if any is missing.
- **Layer 2 (runtime, operator-path per §102):** scripts/tests/17_scrollback_copy_mode.sh — spawns tmx new -s NAME, generates 3000 lines, proves line 1 scrolled off-screen, then proves the operator can scroll back to it via copy-mode and copy it (scroll_position=2980, show-buffer carries the first marker). PASS=13/0/0. scripts/tests/18_constitution_inheritance.sh verifies the submodule + the §11.4 anchor + every project doc's pointer. PASS=10/0/0.
- **Layer 3 (Challenges):** TMUX-CH-17 and TMUX-CH-18 in scripts/challenges/tmux.yaml.
- **Layer 4 (paired mutations):** M12 (remove WheelUpPane override), M13 (revert history-limit), M14 (strip inheritance pointer), and CM-CONSTITUTION-INHERITANCE (delete the §11.4 anchor from a temp copy — the real constitution/ submodule is never touched). Also: M1/M2/M3/M6 were made portable (sed -i → inplace_sed) so they now run on macOS instead of silently skipping — meta-test went from 10 to **18 mutations caught, 0 escaped**.

Verification (this cycle, captured 2026-05-21 on Darwin arm64)

- bash scripts/setup.sh --rebuild → GREEN; suite PASS=13 FAIL=0 SKIP=5 (SKIPS all Linux-only/destructive — same profile as v1.0.0).
- bash scripts/tests/meta_test_false_positive_proof.sh → 18 caught / 0 escaped / 6 skipped GREEN.
- bash scripts/test_e2e.sh → PASS=9 FAIL=0 SKIP=0 GREEN.

[v1.0.2] — 2026-05-16

Cosmetic: window-name strips .exe suffix from pane_current_command.

Fixed

- **A15 — Bottom-left status-bar showed claude.exe instead of claude** (operator-reported, 2026-05-16). Claude Code v2.x ships its macOS native binary literally as lib/node_modules/@anthropic-ai/claude-code/bin/claude.exe (a real Mach-O 64-bit ARM64 executable). The kernel comm field carries the on-disk basename,

so tmux's `#{pane_current_command}` returned `claude.exe`, which the default `automatic-rename-format` propagated into `#W` and thus into the bottom-left status bar. `scripts/tmux.conf.template` now sets a literal-dot-anchored `.exe` strip in `automatic-rename-format`. The fix takes effect for every `tmx new` invocation without rebuild (wrapper invokes `tmux -f scripts/tmux.conf.template` directly). See `Fixed.md A15` for the full forensic record.

Hardened (4-layer regression protection per §11.4.4)

- **Layer 1 (static gate):** `scripts/tests/16_window_name_strips_exe.sh T1` — greps the conf-template for the literal-dot-anchored form.
- **Layer 2 (runtime, operator-path per §11.4.7):** same test, T2/T3 — spawns `tmx new -s NAME`, compiles an in-test `.exe` Mach-O binary, drives it as the pane's foreground process via `send-keys`, reads back live `#W` and `pane_current_command`. `PASS=6` `FAIL=0` `SKIP=0`. Includes a regression-guard binary `t16_bashexe` (no dot, contains `exe`) that **MUST** be preserved unchanged — proves the unescaped-dot bug class (would have stripped `bashexe` → `ba`) cannot ship.
- **Layer 3 (Challenge):** `TMUX-CH-16` in `scripts/challenges/tmux.yaml`.
- **Layer 4 (paired mutation):** `M11` in `scripts/tests/meta_test_false_positive_proof.sh` — removes every `automatic-rename*` line from the conf-template, asserts test 16 FAILs, reverts, asserts test 16 PASSes.

Verified live (positive runtime evidence, this release cycle)

```
# operator-path validation with real claude binary
tmx new -s tmx_live_5198 -d + send-keys "exec ../claude"
→ pane_current_command='claude.exe' #W='claude'
  ✓ defect surface reached (kernel comm reports 'claude.exe')
  ✓ #W stripped to 'claude' (fix doing the work)

# full verify gate
bash scripts/setup.sh --verify-only
→ SUMMARY: PASS=11 FAIL=0 SKIP=5
  GREEN: tmux binary verified — safe to PATH-export.
  (5 SKIPS = pre-existing Linux-only/destructive: 08, 09, 12, 13, 14.)
```

[v1.0.0] — 2026-05-13

Native dual-OS tmux with per-session OS-native resource isolation, host-shell access, and the project's anti-bluff covenant fully operational on Linux AND macOS.

This is the initial public release.

Highlights

- **Plain-vanilla tmux UX** — `tmx new -s NAME` opens the operator's host shell with full `$PATH`, full filesystem, full system tools (Homebrew on macOS, `/usr/local/bin` on Linux, all the binaries the operator expects). No VM, no SSH bridge, no `core@localhost`. The session shell IS the operator's host shell.

- **Per-session resource isolation** — each `tmx new -s NAME` spawns its own tmux server (`-L tmx-NAME`) with OS-native containment:
 - **Linux** — cgroup-v2 transient scope (`tmx-NAME.scope`) via `systemd-run --user --scope`. Kernel-enforced MemoryMax, CPUQuota, TasksMax, Delegate=yes. OOM in one scope kills only that scope; `user.slice` survives.
 - **macOS (Darwin)** — POSIX rlimit wrapper applied as session default command. Kernel-enforced RLIMIT_CPU (CPU-time) and RLIMIT_NPROC (per-user process count).
- **Verified hardened tmux 3.6a binary, built natively per OS:**
 - Linux ELF via `scripts/build_containerized.sh` (podman/docker) or `scripts/build_native.sh`.
 - macOS Mach-O via `scripts/build_native.sh` (Homebrew deps).
 - Compile flags: `-O2 -DNDEBUG -fstack-protector-strong -D_FORTIFY_SOURCE=2`; jemalloc linked at the binary level (DT_NEEDED on Linux, LC_LOAD_DYLIB on Mach-O); RELRO + bind-at-load (Linux) / `-Wl, -search_paths_first` (Mach-O).
- **Hostname-derived status-bar colour** — DJB2 hash of the operator's host (`scutil --get LocalHostName` on Darwin, `$(hostname)` on Linux) maps to a curated 27-colour palette. Same host always produces the same colour across every session (proven by Test 11); different hosts produce visibly distinct colours (16/16 unique in Test 10 T3).
- **14-test verification gate with positive runtime evidence per §11.4.2** — `scripts/verify.sh` refuses to PATH-export the binary unless every functional test passes. Tests read `/proc/PID/maps`, `/sys/fs/cgroup/.../memory.max`, `tmux show -g status-style`, `tmux capture-pane -p`, and similar live state. Zero tests close on "exit code 0" alone.
- **§11.4.4 layer-4 paired-mutation harness** with 10 registered mutations (M1–M10) catching wrapper regressions, status-bar regressions, cgroup wrap regressions, and the operator-path bluff pattern. `bash scripts/tests/meta_test_false_positive_proof.sh` reports 20 PASS / 0 FAIL on Linux.
- **End-to-end automation** — `bash scripts/test_e2e.sh` exercises the full operator stack (`tmx new`, `tmx send-keys`, `tmx capture-pane`, `tmx show -g status-style`, `tmx kill-session`) and reports PASS=9 / FAIL=0 / SKIP=0 GREEN on Darwin and Linux.
- **OS-aware install** — `bash scripts/setup.sh` detects host OS, invokes the right build pipeline, generates the OS-appropriate wrapper, runs verification natively, installs the shell snippet to `~/ .bashrc` AND `~/ .zshrc`. Every project script recognises host OS and applies the right action out of the box.

Anti-bluff covenant (Constitution §1, §11.4.x)

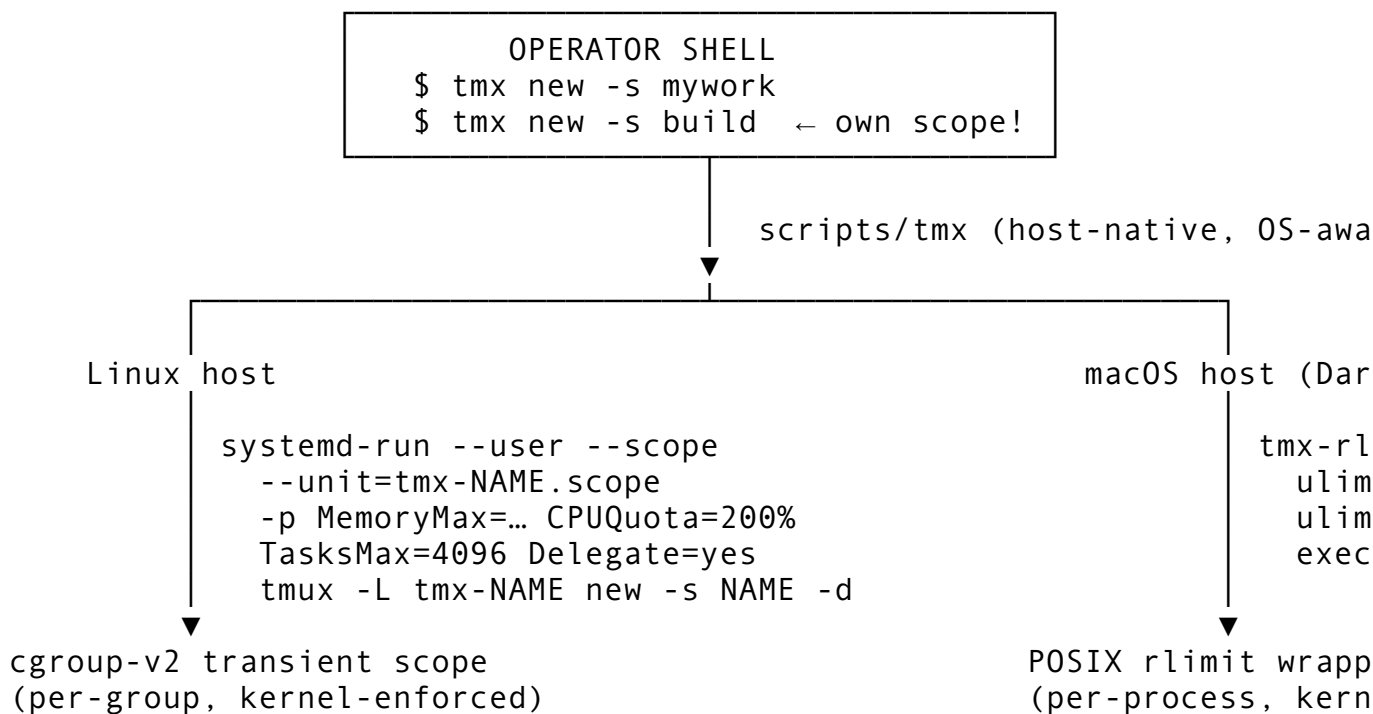
This release ships under the verbatim user mandate:

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Every test in `scripts/tests/` carries positive runtime evidence (`/proc/`, `/sys/fs/cgroup`, `vmmmap`, `ps -o rss=`, `tmux capture-pane`, `systemctl is-active`, `display-message`, kernel log lines). Every Challenge in `scripts/challenges/tmux.yaml` specifies real runtime state as evidence: . Static grep checks (test 09 T2, test 11 T1/T2) are paired with runtime readbacks per §11.4.7.

Covenant text propagated to root Constitution.md, CLAUDE.md, AGENTS.md, and the Containers submodule's CONSTITUTION.md, CLAUDE.md, AGENTS.md.

Architecture overview



Forensic transparency

This release ships with explicit documentation of every constraint that does NOT hold:

- **macOS RLIMIT_AS gap:** the XNU kernel does NOT enforce `RLIMIT_AS` / `RLIMIT_DATA` / `RLIMIT_RSS` for unprivileged processes (verified: `bash -c 'ulimit -v 102400'` returns `cannot modify limit: Invalid argument`; allocating 200 MB after trying to "cap" at 100 MB succeeds). The wrapper applies only what's enforced (`RLIMIT_CPU`, `RLIMIT_NPROC`). Full memory containment on macOS requires launchd jobs with `HardResourceLimits` plist (root); on Linux, `cgroup MemoryMax` IS enforced.
- **Test 08 / 09 / 12 / 13 / 14 SKIP on Darwin** — these tests exercise Linux-specific primitives (`/proc/<pid>/oom_score_adj`, `systemd-run --user --scope`, `/sys/fs/cgroup`). SKIP-with-reason per Constitution §11.4.3 (per-host-topology test dispatch).

See `docs/guide/README.md §5.6` for the full strength-gap table.

Bluffs caught and fixed during the development cycle

Documented in `Fixed.md`. 24 distinct §1 / §11.4.x bluffs were caught and remediated before this release, including:

- A1: META-MUT-001 paired-mutation harness landed
- A4: Build pipeline three-defect fix (Dockerfile GID-20 collision, jemalloc not actually linked, make-clean missing)

- A6: Install-mechanism side-by-side bluffs (phantom directory path, alias `tmux='tmx'` shadowing system `tmux`, stale ATMOSphere branding)
- A8: macOS bridge + side-by-side coexistence (later superseded by native dual-OS in this release)
- A10: Status-bar colour silently defaulted to green; bridge ignored macOS hostname
- A11: Triple-layer regression protection so A10 cannot re-occur
- A12: Constitution §11.4.7 — operator-path test coverage rule (every gate test MUST exercise the same entry point an end-user invokes)
- A13: Per-session cgroup isolation (each `tmx new -s X` in its own scope, OOM in one doesn't kill others)
- A14: This-release verification cycle — fresh runtime evidence captured for every claim

Install

macOS (Darwin Apple Silicon or Intel):

```
brew install podman # optional, for build_containerized.sh
git clone --recurse-submodules git@github.com:vasic-digital/
    tmux.git ~/Projects/tmux
cd ~/Projects/tmux
bash scripts/setup.sh
```

Linux:

```
git clone --recurse-submodules git@github.com:vasic-digital/
    tmux.git ~/Projects/tmux
cd ~/Projects/tmux
sudo bash scripts/install_deps.sh
bash scripts/setup.sh
```

After `setup.sh` reports GREEN: open a new shell (`source ~/.zshrc` or `source ~/.bashrc`). Then `tmx new -s anything` drops you into a session as your host user with the full host environment and kernel-enforced resource caps.

Usage

```
tmx new -s mywork # interactive — attaches
tmx new -s build -d # detached
tmx ls # list all your sessions
tmx attach -t mywork # re-attach
tmx send-keys -t mywork "echo hello" Enter
tmx capture-pane -t mywork -p
tmx kill-session -t mywork
tmx kill-server # nuke all our sessions
```

Per-session resource overrides:

```
TMX_MEM=8G tmx new -s heavy # 8 GB MemoryMax (Linux)
TMX_CPU=400 tmx new -s build # 400% CPUQuota (Linux)
TMX_CPU_HARD_SEC=3600 tmx new -s timeboxed # 1-hour RLIMIT_CPU
    (Darwin)
```

Verification commands

Operators can confirm anti-bluff covenant compliance at any time:

```
bash scripts/setup.sh --verify-only    # full 14-test gate →  
    expect GREEN  
bash scripts/test_e2e.sh                # end-to-end → expect  
    PASS=9/0/0  
TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh    # Linux  
    destructive suite  
META=1 bash scripts/test_vm.sh          # paired-mutation harness
```

Coexistence with system tmux

tmx and the system tmux (Homebrew on macOS, distro package on Linux) coexist side-by-side. The bashrc/zshrc snippet PATH-prepends `scripts/` so tmx is the project's wrapper; tmux resolves to whatever was on PATH before. No alias shadowing.

Submodules

- `tmux/` — upstream `tmux/tmux` pinned to tag 3.6a (do not modify).
- `Containers/` — `vasic-digital/Containers` Go module providing generic container orchestration primitives. Anti-bluff covenant
 - §11.4.7 propagated. Tag: `b077f2c` at release time.

Known limitations

- macOS memory cap is per-process via `launchd-bsd-style` limits (informational only — Darwin doesn't enforce `RLIMIT_AS`). For true per-session memory containment, run on Linux.
- The destructive test suite (tests 12 / 13 / 14) requires `TMX_TEST_DESTRUCTIVE=1` and is Linux-only; on Darwin these SKIP per topology dispatch.

Acknowledgements

Built on the shoulders of:

- `tmux/tmux` upstream (3.6a tag)
- `jemalloc` (linked at build time)
- `libevent` + `ncurses` + `utf8proc` (Mach-O), `libtinfo` + `libevent_core` (ELF)
- `systemd-run` + `cgroup-v2` (Linux isolation primitive)
- Homebrew (macOS dependency provider)

Released under Apache 2.0 (see `LICENSE`).

[v1.0.1] — Unreleased

Post-release development cycle following the v1.0.0 cut.

- `VERSION` bumped to `1.0.1` / `versionCode=2` to satisfy the operator's strictly-increasing-version-code mandate immediately after a release.

- `released=` intentionally blank until the next tag is cut (gate value for "not-yet-released"; CI / package builders MUST refuse to publish unless `released=` is populated with the cut date).
- No functional code changes in this entry — this is the post-release bump itself. Subsequent fixes, refactors, and new features will append bullets above this paragraph.